# Section 2.8 — Rational Functions

Math 103: College Algebra

**Identifying Rational Expressions**

Identify each expression as rational or not.

1. $\dfrac{2x^2-5x+3}{24x^3-5}$
2. $\dfrac{wx^2y}{\sqrt{25x}}$
3. $\dfrac{2x-4}{x+\sqrt{2}}$
4. $2x^2-4x+5z$

**Negative Exponents**

Rewrite each expression involving negative exponents as a rational expression.

5. $2x^{-4}y^2z^{-1}$
6. $\frac{2}{3}u^{-2}w^5$
7. $12a^{-1}b^{-7}$
8. $\frac{1}{4}j^3k^{-3}$

**Rational Functions**

For each pair of functions $f$ and $g$, write the function $\left(\frac{f}{g}\right)(x)$ as a rational function, and find its domain. **Do not simplify** your rational function.

9. $f(x)=2x^2-5x+3, \quad g(x)=2x^2-5x-7$
10. $f(x)=2x-3, \quad g(x)=4x^2-9$

**Evaluating Rational Functions**

Let $h(x)=\dfrac{2x^2-5x+3}{x^3-3x^2+3x-3}$. Find

11. $h(2)$
12. $h(-3)$
13. $h(\frac{1}{2})$
14. $h(1)$

Let $s(x)=\dfrac{x^2-5x-11}{2x^2+5}$

15. $s(7)$
16. $s(-1)$
17. $s(1.5)$
18. $s(0)$

**Simplifying Expressions**

Simplify each expression. You may leave your answer in factored form.

19. $\dfrac{14h^2k}{21h^3}$
20. $-\dfrac{5a^3b}{a^2b^4}$
21. $-\dfrac{5x^2y}{10x^3y}$
22. $\dfrac{2x^2yu^4}{8x^7u^8z^2}$
23. $\dfrac{3(m-2)}{5(m-2)}$
24. $\dfrac{(x+4)(3x-5)(x^2+1)}{(x-4)(3x-5)(x+1)^2}$
25. $\dfrac{5-b}{b-5}$
26. $\dfrac{10x-20}{15x-30}$
27. $\dfrac{3x-5}{10x-6x^2}$
28. $\dfrac{6b^2+b-12}{10b^2+13b-3}$
29. $\dfrac{px+py+qx+qy}{mx-nx+my-ny}$
30. $\dfrac{a^2-b^2}{a-b}$
31. $\dfrac{x^2-y^2}{x^3-y^3}$
32. $\dfrac{x^3-16x}{x^3+6x^2+8x}$
33. $\dfrac{8n^3+20n^2-20n-50}{4n^2+12n+5}$
34. $\dfrac{x^3-8}{x^3+2x^2+4x}$

**Simplifying Functions**

Simplify each function. Note the domain restrictions, if any. You may leave your answer in factored form.

35. $f(x)=\dfrac{3x}{3x^2-9x}$
36. $g(x)=\dfrac{x-1}{4x^2+4x-8}$
37. $h(x)=\dfrac{3x-7}{3x^2-4x-7}$
38. $k(x)=\dfrac{x^2-9}{2x^2-7x+3}$
39. $s(x)=\dfrac{4x^2+5x-6}{x^3+8}$
40. $t(x)=\dfrac{2x^2+7x+6}{18+9x-8x^2-4x^3}$